北京大學
PEKING UNIVERSITY

第一章 量子理论

3. 平动、振动和转动

问题一　平动

1. 一维自由粒子

势能为零，写出薛定谔方程 $$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2}=E\psi(x)$$

通解为 $$\psi_k(x)=Ae^{ikx}+Be^{-ikx} \qquad E_k=\frac{k^2\hbar^2}{2m}$$

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\overbrace{(Ae^{ikx}+Be^{-ikx})}^{\psi_k(x)}=\overbrace{\frac{k^2\hbar^2}{2m}}^{E_k}\overbrace{(Ae^{ikx}+Be^{-ikx})}^{\psi_k(x)}$$

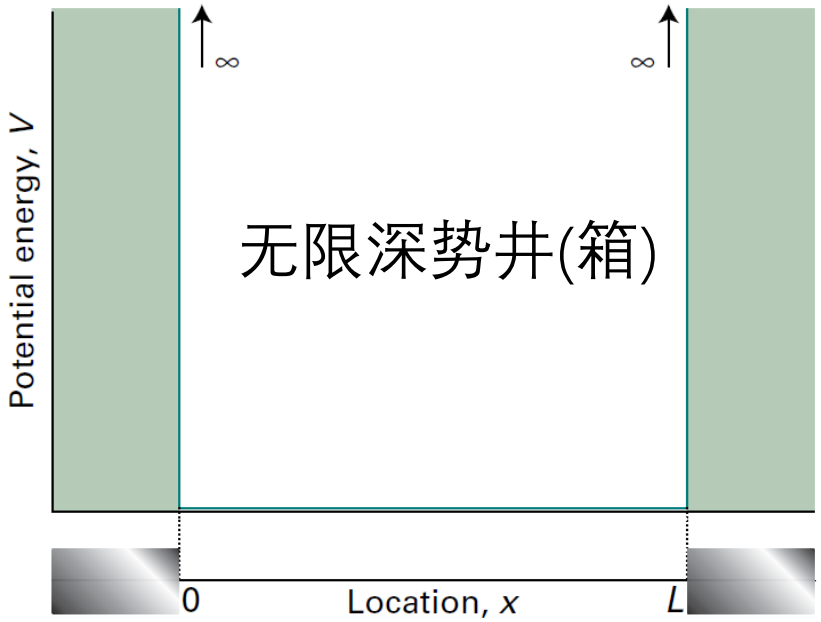
注意：能量没有量子化，可以连续变化，原因是没有势能约束

本征态无法归一化，是两列平面波的叠加

$$\psi_k(x)=\underbrace{A\,e^{+ikx}}_{\text{Particle with linear momentum } +k\hbar}+\underbrace{B\,e^{-ikx}}_{\text{Particle with linear momentum } -k\hbar}$$

思考：三维自由粒子的本征值和本征态 $$-\frac{\hbar^2}{2m}\nabla^2\psi=E\psi$$

2. 一维束缚粒子



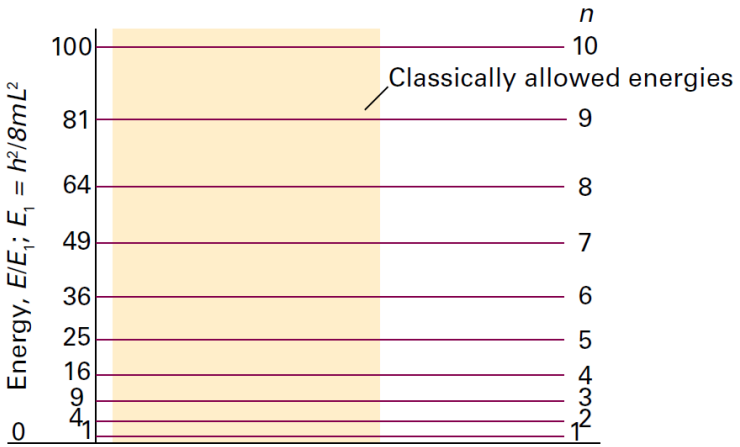
井内 $\psi_k(x) = C\sin kx + D\cos kx$

井外 For $x<0$ and $x>L$, $\psi_k(x)=0$

边界连续性条件 $\psi_k(0)=0$ and $\psi_k(L)=0$

$$\Rightarrow \quad D=0 \qquad k=\frac{n\pi}{L}, \quad n=1,2,\cdots$$

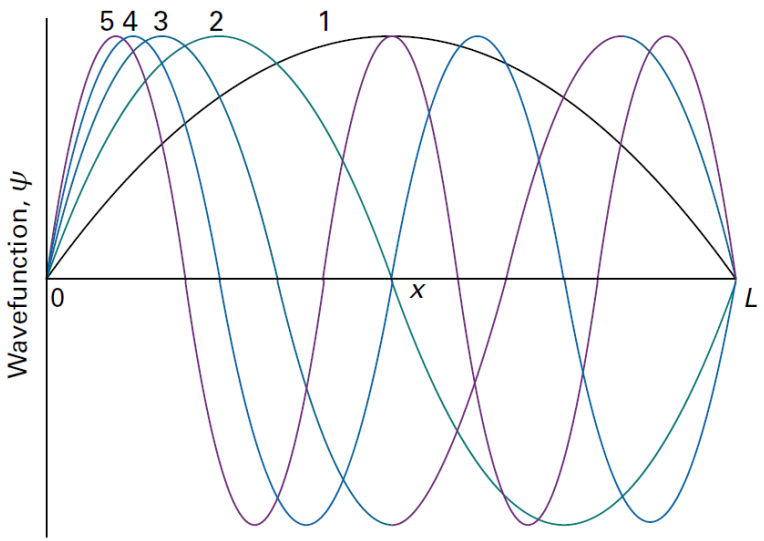
约束导致量子化!

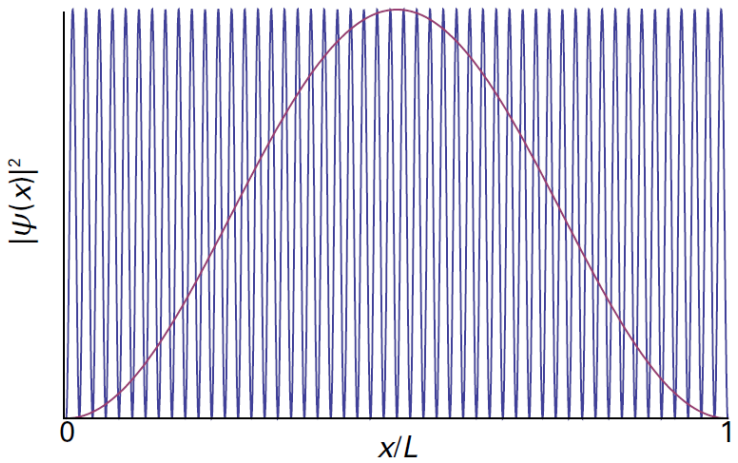


$$\psi_n(x) = \left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{n\pi x}{L}\right) \qquad E_n = \frac{n^2h^2}{8mL^2} \qquad n=1,\ 2,\ldots$$

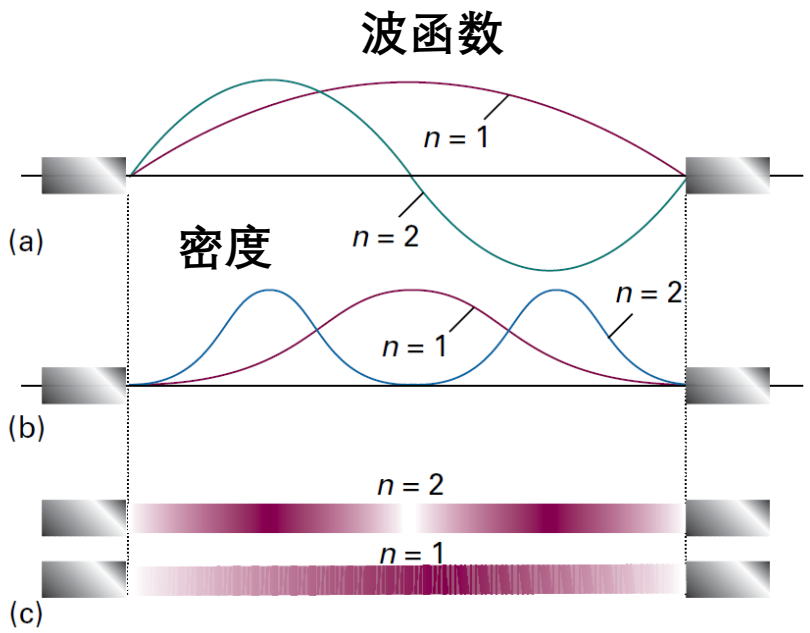
思考：如何解释波函数在边界处导数不连续?

本征波函数的性质





n = 50



零点能（zero-point energy)

$$E_1 = \frac{h^2}{8mL^2}$$

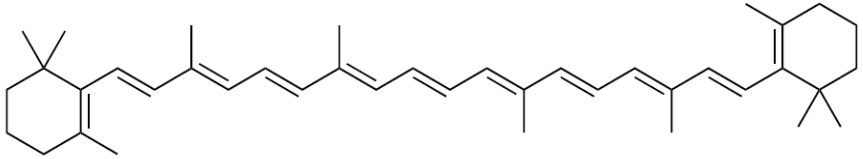
零点能的解读，参考教材

高量子数，波函数在各处出现的概率密度变得均一，逼近经典行为

无限深势井的应用场景:

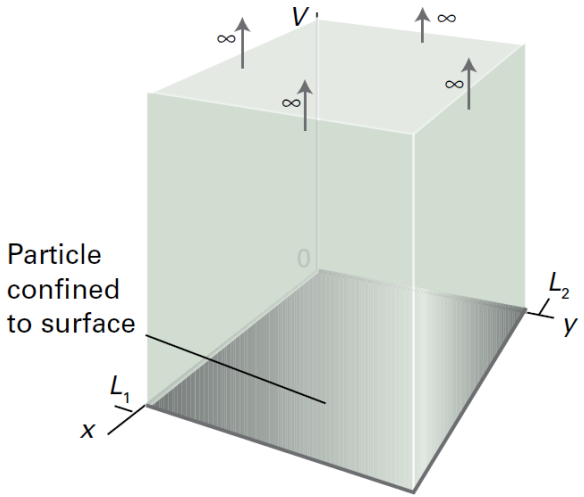
1. 实验室反应釜中，稀薄气体分子的运动（三维势箱）

2. 共轭分子的大Π键，预测激发能（只能定性，定量不太准）



1 β-Carotene

多维势箱



不同维度独立，可分离变量

$$\overbrace{-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}}^{\hat{H}_x}\psi-\overbrace{\frac{\hbar^2}{2m}\frac{\partial^2}{\partial y^2}}^{\hat{H}_y}\psi=E\psi$$

$$\psi(x,y)=X(x)Y(y)$$

$$\overbrace{\frac{1}{X(x)}\hat{H}_xX(x)}^{\text{Depends only on } x}+\overbrace{\frac{1}{Y(y)}\hat{H}_yY(y)}^{\text{Depends only on } y}=\overbrace{E}^{\text{A constant}}$$

二维势箱的解

$$\frac{1}{X(x)}\hat{H}_x X(x) = E_X,\ \text{so}\ \ \hat{H}_x X(x) = E_X X(x)$$

$$\frac{1}{Y(y)}\hat{H}_y Y(y) = E_Y,\ \text{so}\ \hat{H}_y Y(y) = E_Y Y(y)$$

分离变量后，转换为一维势箱问题

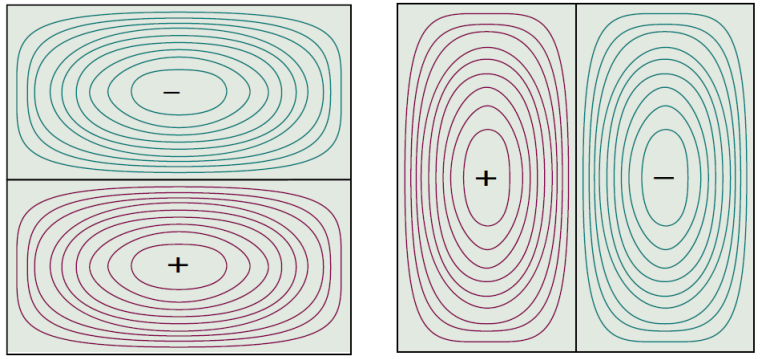
$$X_{n_1}(x) = \left(\frac{2}{L_1}\right)^{1/2} \sin\left(\frac{n_1 \pi x}{L_1}\right) \qquad E_{X,n_1} = \frac{n_1^2 h^2}{8mL_1^2}$$

$$Y_{n_2}(y) = \left(\frac{2}{L_2}\right)^{1/2} \sin\left(\frac{n_2 \pi y}{L_2}\right) \qquad E_{Y,n_2} = \frac{n_2^2 h^2}{8mL_2^2}$$

最后的解依赖于两个量子数

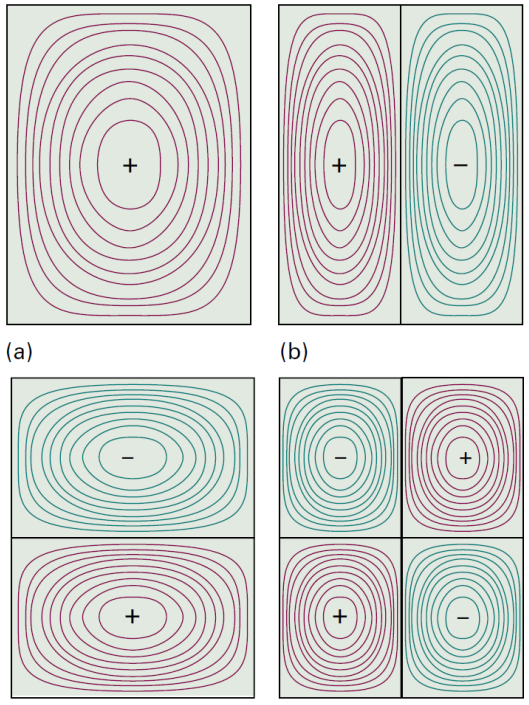
$$\psi_{n_1,n_2}(x,y) = \frac{2}{(L_1 L_2)^{1/2}} \sin\left(\frac{n_1 \pi x}{L_1}\right) \sin\left(\frac{n_2 \pi y}{L_2}\right)$$

$$E_{n_1,n_2} = \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2}\right)\frac{h^2}{8m}$$



(a) (b)

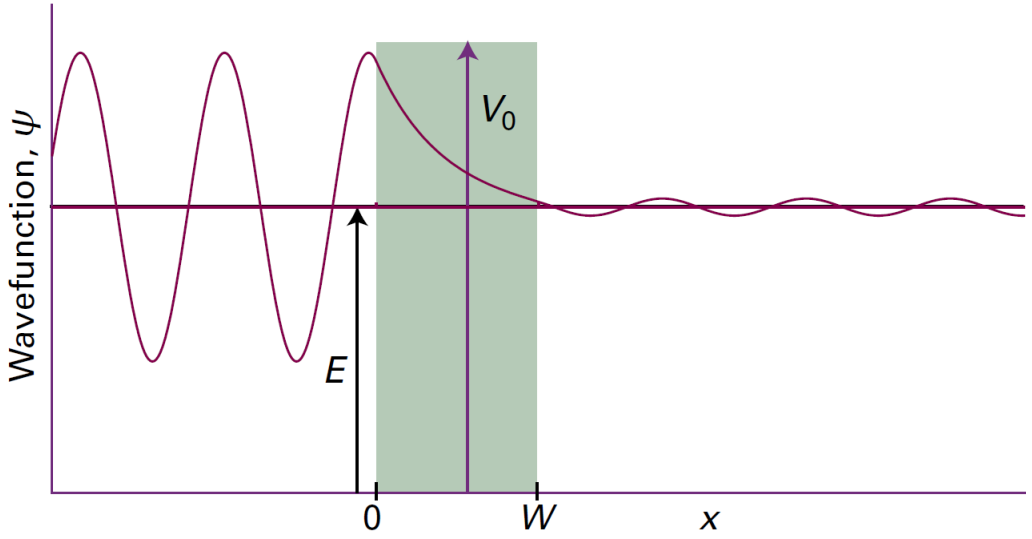
方势井的简并态



(a) (b) (c) (d)

长方势井
波函数等值面

3. 一维有限宽有限深势垒



该问题本征态分多种情况

这里只考虑势垒左边是平面波

解读为左边的入射波

注意：并非含时问题，可理解为达到“平衡”后的定态

求解时分三段考虑：

首先假定 $E < V_0$

$$\psi = Ae^{ikx} + Be^{-ikx} \qquad k\hbar = (2mE)^{1/2}$$

Wavefunction to left of barrier

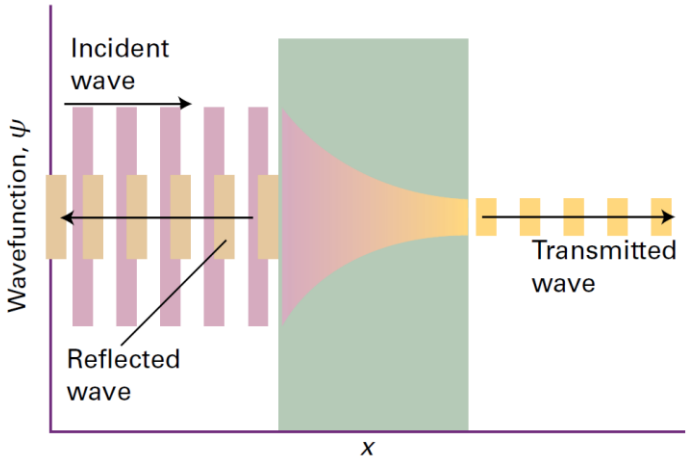
$$\psi = Ce^{\kappa x} + De^{-\kappa x} \qquad \kappa\hbar = \{2m(V_0 - E)\}^{1/2}$$

Wavefunction inside barrier

$$\psi = A'e^{ikx} \qquad k\hbar = (2mE)^{1/2}$$

Wavefunction to right of barrier

势垒右侧基于物理，不考虑反射波



求解系数，考虑边界连续性条件：

$$\text{at } x=0\text{: } A+B=C+D \qquad \text{at } x=W\text{: } Ce^{\kappa W}+De^{-\kappa W}=A'e^{ikW}$$

边界导数连续性条件：

$$\text{at } x=0\text{: } ikA-ikB=\kappa C-\kappa D$$
$$\text{at } x=W\text{: } \kappa Ce^{\kappa W}-\kappa De^{-\kappa W}=ikA'e^{ikW}$$

四个方程，五个未知系数，可定出比例关系

定义透射概率 $T=\dfrac{|A'|^2}{|A|^2}$

计算得 $T=\left\{1+\dfrac{(e^{\kappa W}-e^{-\kappa W})^2}{16\varepsilon(1-\varepsilon)}\right\}^{-1}$

其中 $\varepsilon = E/V_0$

$$\psi=Ae^{ikx}+Be^{-ikx} \qquad k\hbar=(2mE)^{1/2}$$

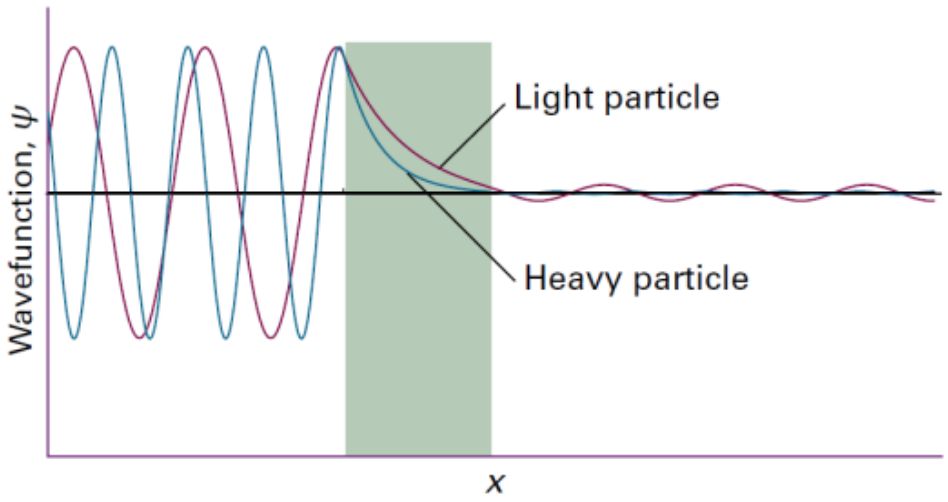
Wavefunction to left of barrier

$$\psi=Ce^{\kappa x}+De^{-\kappa x} \qquad \kappa\hbar=\{2m(V_0-E)\}^{1/2}$$

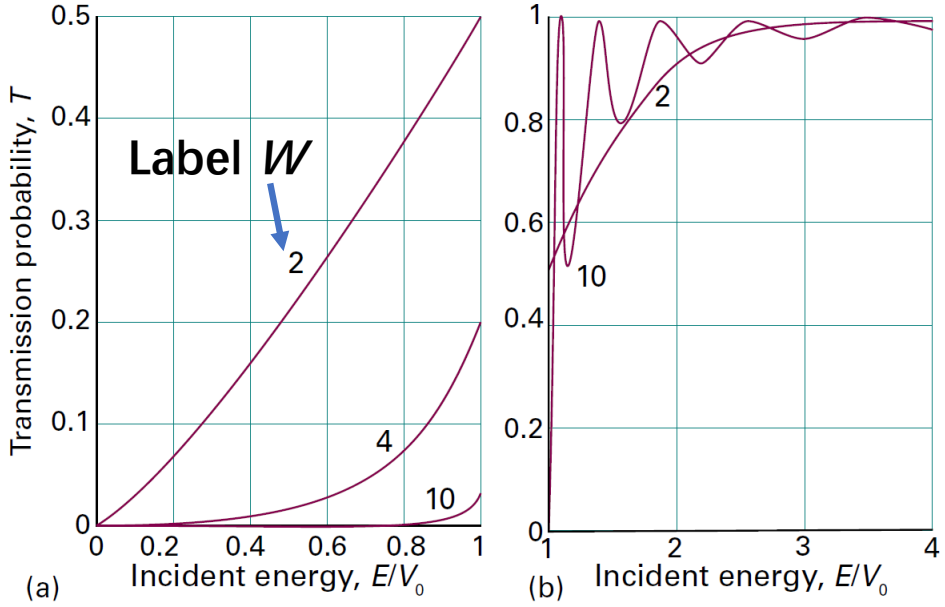
Wavefunction inside barrier

$$\psi=A'e^{ikx} \qquad k\hbar=(2mE)^{1/2}$$

Wavefunction to right of barrier



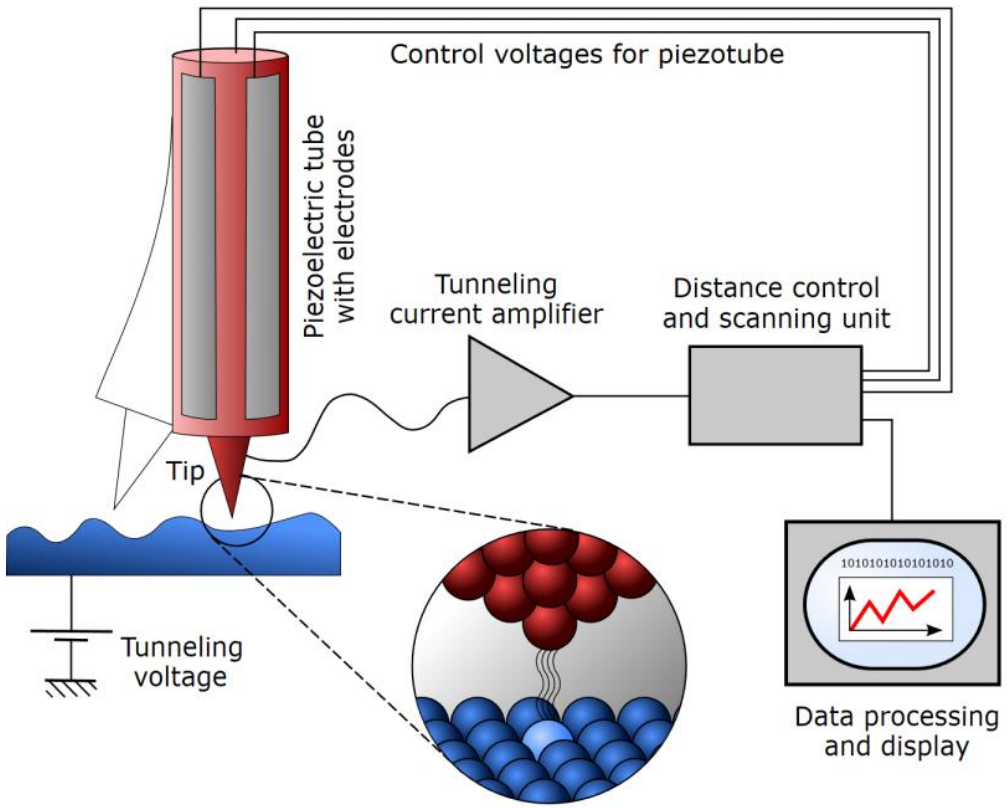
隧穿概率与质量有关，
质量越大，越难隧穿



对宽势垒， $\kappa W >> 1$

$$T \approx 16\varepsilon(1-\varepsilon)e^{-2\kappa W}$$

隧穿概率指数衰减，STM的原理



From wikipedia

量子图像：即使能量<势垒，也有一定穿越概率

即使能量>势垒，也不一定100%穿越！

经典图像：如果能量<势垒，只能出现在势垒一侧

如果能量>势垒，则100%可以穿越

注意：由于没有束缚，能量可以连续取值

思考：E>V如何求解

电子密度与密度算符

$$\hat{\rho}(x) = \delta(x - x_1)$$

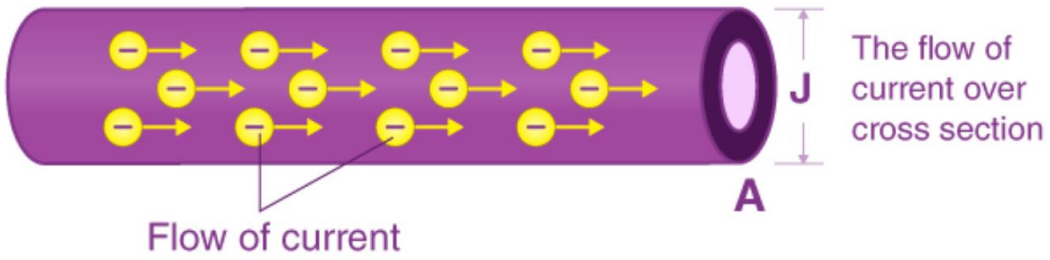
$$\rho(x) = |\psi(x)|^2 = \int \psi^*(x_1)\delta(x - x_1)\psi(x_1)dx_1$$

$$= \langle\psi|\hat{\rho}(x)|\psi\rangle$$

电子密度等于密度算符的期望值

电流密度

$$j(x) = \frac{N}{A\Delta t} = \frac{\rho V}{A\Delta t} = \rho v$$



$$\hat{j}(x) = \frac{1}{2}\left\{\hat{\rho}(x), \frac{\hat{p}_x}{m}\right\}$$

$$j(x) = \langle\psi|\hat{j}(x)|\psi\rangle$$

电流密度等于电流密度算符的期望值

计算电流密度

$$j(x) = \langle\psi|\hat{j}(x)|\psi\rangle \ = \frac{1}{2m}\langle\psi|\hat{\rho}(x)\hat{p}_x + \hat{p}_x\hat{\rho}(x)|\psi\rangle$$

$$\langle\psi|\hat{\rho}(x)\hat{p}_x|\psi\rangle = \int dx_1\psi^*(x_1)\delta(x-x_1)(-i\hbar)\frac{d\psi(x_1)}{dx_1}$$

$$= -i\hbar\psi^*(x)\frac{d\psi(x)}{dx}$$

$$\langle\psi|\hat{p}_x\hat{\rho}(x)|\psi\rangle = \langle\hat{p}_x\psi|\hat{\rho}(x)|\psi\rangle \ = \langle\hat{\rho}(x)\psi|\hat{p}_x\psi\rangle^*$$

两项互为复共轭

$$= \langle\psi|\hat{\rho}(x)\hat{p}_x|\psi\rangle^* = i\hbar\psi(x)\frac{d\psi^*(x)}{dx}$$

$$\boxed{\begin{aligned} j(x) = \langle\psi|\hat{j}(x)|\psi\rangle &= \frac{-i\hbar}{2m}\left(\psi^*(x)\frac{d\psi(x)}{dx} - \psi(x)\frac{d\psi^*(x)}{dx}\right) \\ &= \frac{\hbar}{m}\mathrm{Im}\left\{\psi^*(x)\frac{d\psi(x)}{dx}\right\} \end{aligned}}$$

束缚态一般电流密度为零　　波函数取实数

非束缚态电流密度可以不为零　　波函数取复数

正向传播的平面波　　$\psi_k(x) = Ae^{ikx}$

$$j(x) = \frac{\hbar}{m}\mathrm{Im}\left\{\psi^*(x)\frac{d\psi(x)}{dx}\right\} = \frac{|A|^2 k\hbar}{m}$$

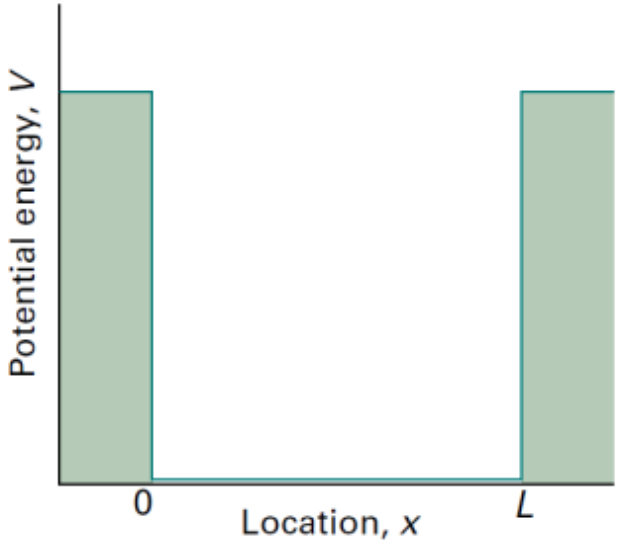
负向传播的平面波　　$\psi_{-k}(x) = Be^{-ikx}$

$$j(x) = \frac{\hbar}{m}\mathrm{Im}\left\{\psi^*(x)\frac{d\psi(x)}{dx}\right\} = \frac{-|B|^2 k\hbar}{m}$$

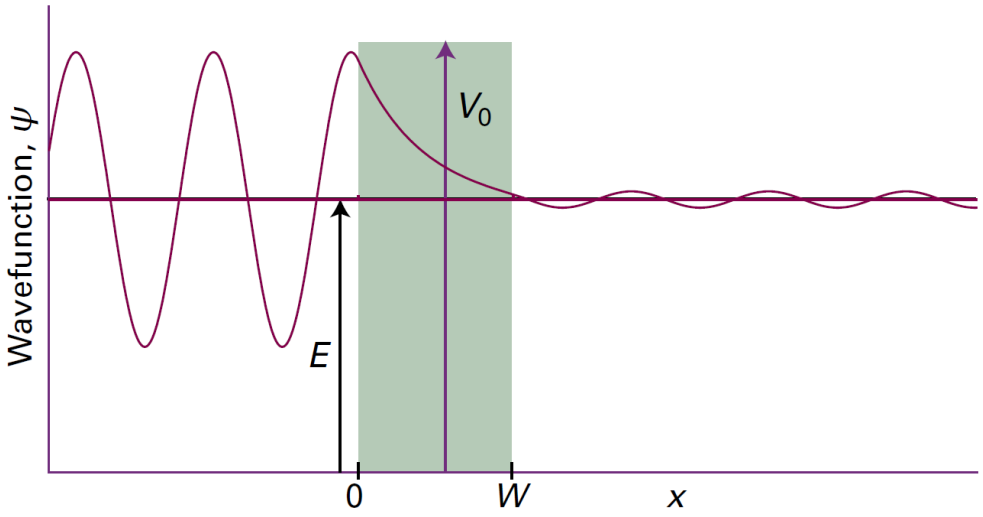
隧穿电流与初始电流之比　　$\frac{I_t}{I_0} = \frac{|A'|^2}{|A|^2} = T$　　等于透射率

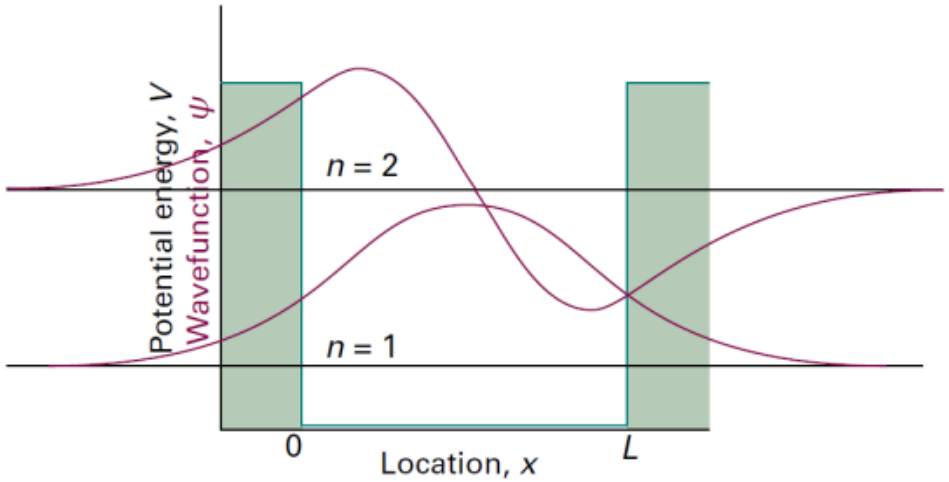
4. 一维有限宽有限深势井

势井：有束缚态，能量量子化



势垒：非束缚态，能量连续





束缚态个数N

$$N-1<\frac{(8mV_0L)^{1/2}}{h}<N$$

有限深(高)势井(垒)导致波函数可以穿井(垒)

问题二　振动

简谐振子方程

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+\tfrac{1}{2}k_f x^2\psi(x)=E\psi(x)$$

边界条件：波函数在无穷远处为零

结论：

$$E_v=\left(v+\tfrac{1}{2}\right)\hbar\omega \qquad \omega=(k_f/m)^{1/2}$$

v 是振动量子数

振动零点能 $E_0=\tfrac{1}{2}\hbar\omega$

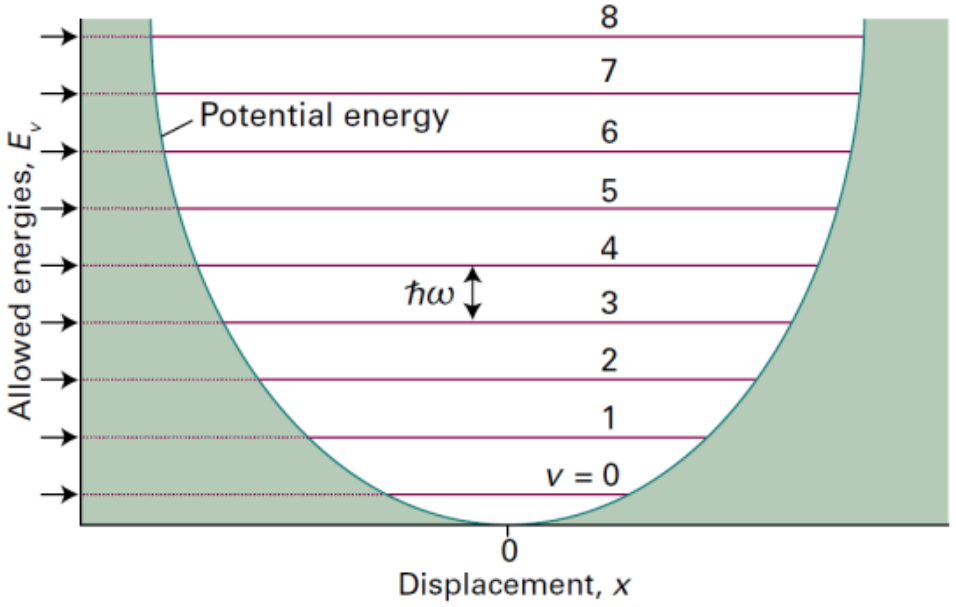
谐振子模型可用来描述
双原子分子的振动

经典运动方程

$$m\ddot{x}=-k_f x$$

$$x=A\sin(\omega t+\theta)$$

注意：量子力学**定态**方程里没有时间！



振动波函数

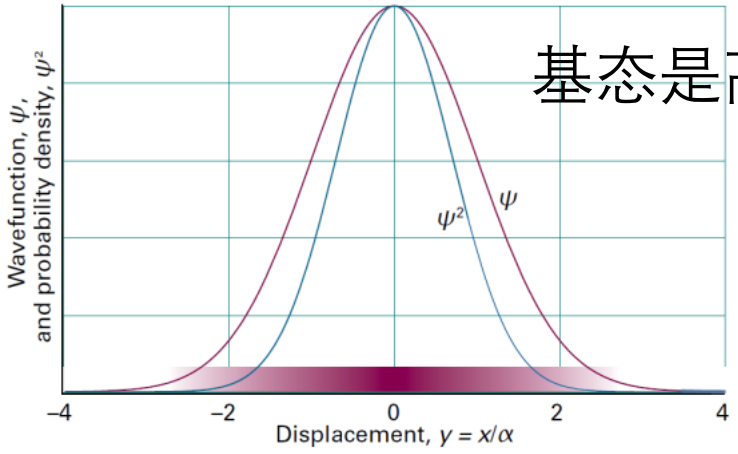
首先，令 $$E=(v+\frac{1}{2})\hbar\omega=(v+\frac{1}{2})\hbar\sqrt{\frac{k_f}{m}}$$

将ψ对E的依赖转化为对v的依赖

利用变量替换，令 $$\psi_v(x)=N_vH_v(y)e^{-y^2/2} \qquad y=\frac{x}{\alpha} \quad \alpha=\left(\frac{\hbar^2}{mk_f}\right)^{1/4}$$

代入方程，得到 $$H_v''-2yH_v'+2vH_v=0$$

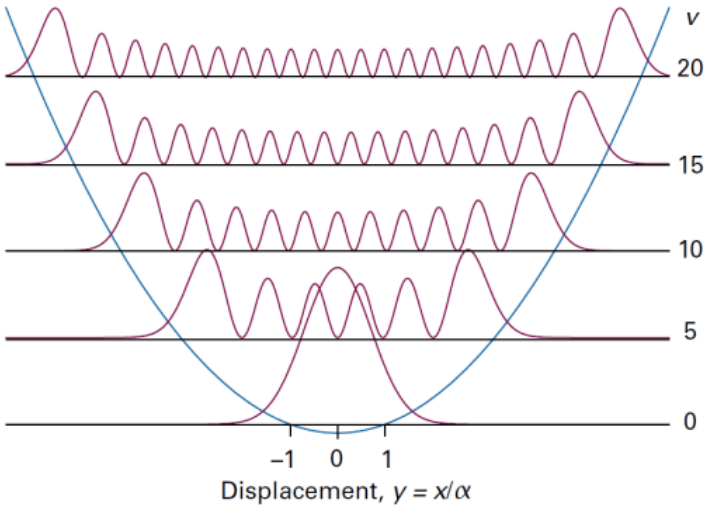
其解为厄密多项式(Hermite polynomials)

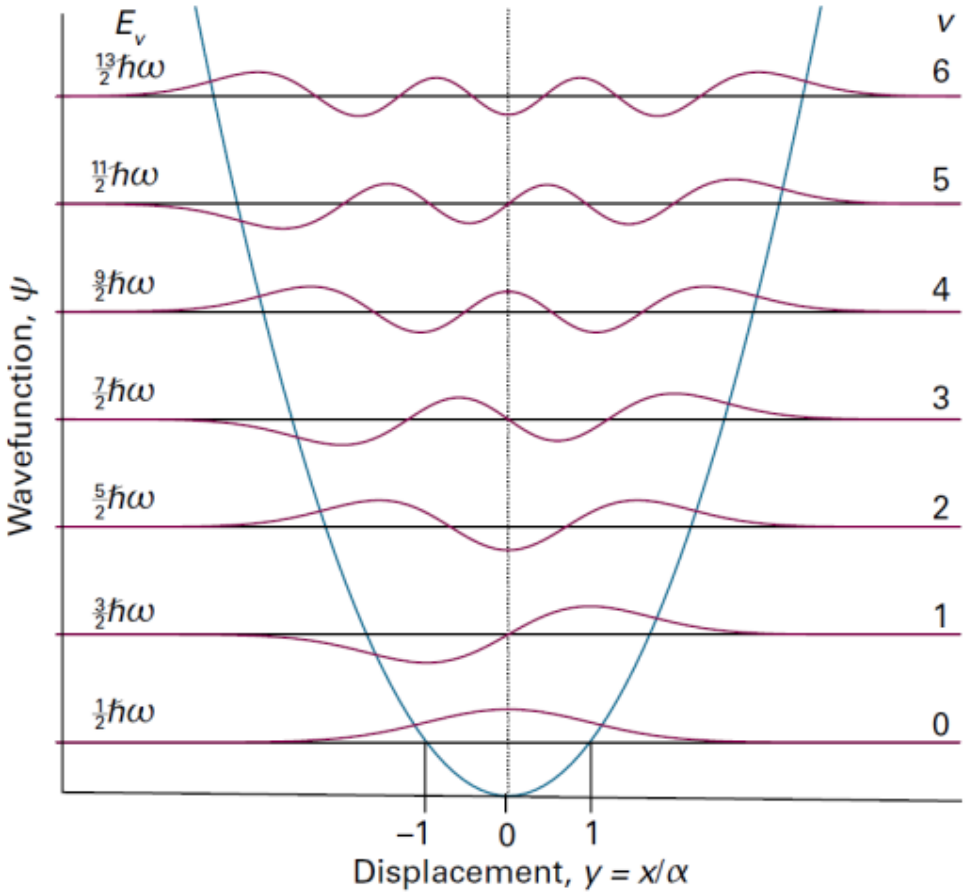


基态是高斯函数

v	$H_v(y)$
0	1
1	$2y$
2	$4y^2-2$
3	$8y^3-12y$
4	$16y^4-48y^2+12$
5	$32y^5-160y^3+120y$
6	$64y^6-480y^4+720y^2-120$

- 谐振子波函数有**奇偶性**，与v的奇偶性一致
- **激发态有节点**，v越大节点越多
- v 越大，振动的范围越大
- 各态能量等间距分布





概率密度

简谐波函数的性质

利用对称性，易证 $\langle x\rangle_v = 0$

位移平方期望值

$$\langle x^2\rangle_v = N_v^2\int_{-\infty}^{\infty}(H_v e^{-y^2/2})x^2(H_v e^{-y^2/2})dx$$

$x=\alpha y \quad dx=\alpha dy$

$$=\alpha^3 N_v^2\int_{-\infty}^{\infty}(H_v e^{-y^2/2})y^2(H_v e^{-y^2/2})dy$$

$$\langle x^2\rangle_v = \alpha^3 N_v^2\int_{-\infty}^{\infty}(H_v e^{-y^2/2})\overbrace{\left\{v(v-1)H_{v-2}+\left(v+\tfrac{1}{2}\right)H_v+\tfrac{1}{4}H_{v+2}\right\}}^{y^2H_v}e^{-y^2/2}dy$$

$$=\alpha^3 N_v^2 v(v-1)\overbrace{\int_{-\infty}^{\infty}H_v H_{v-2}e^{-y^2}dy}^{0}$$

$$+\alpha^3 N_v^2\left(v+\tfrac{1}{2}\right)\overbrace{\int_{-\infty}^{\infty}H_v H_v e^{-y^2}dy}^{\pi^{1/2}2^v v!}+\tfrac{1}{4}\alpha^3 N_v^2\overbrace{\int_{-\infty}^{\infty}H_v H_{v+2}e^{-y^2}dy}^{0}$$

$$=\alpha^3 N_v^2\left(v+\tfrac{1}{2}\right)\pi^{1/2}2^v v!$$

利用厄密多项式
递推关系

$$yH_v = vH_{v-1}+\tfrac{1}{2}H_{v+1}$$

$$y^2H_v = vyH_{v-1}+\tfrac{1}{2}yH_{v+1}$$

$$yH_{v-1} = (v-1)H_{v-2}+\tfrac{1}{2}H_v$$

$$yH_{v+1} = (v+1)H_v+\tfrac{1}{2}H_{v+2}$$

$$y^2H_v = vyH_{v-1}+\tfrac{1}{2}yH_{v+1} = v\{(v-1)H_{v-2}+\tfrac{1}{2}H_v\}$$
$$+\tfrac{1}{2}\{(v+1)H_v+\tfrac{1}{2}H_{v+2}\}$$

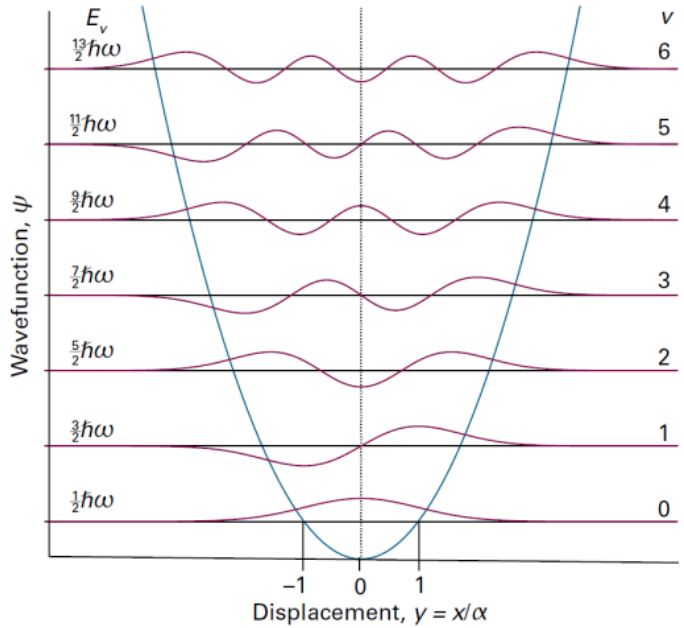
$$= v(v-1)H_{v-2}+\left(v+\tfrac{1}{2}\right)H_v+\tfrac{1}{4}H_{v+2}$$

代入归一化常数得

$$\langle x^2\rangle_v = \frac{\alpha^3\left(v+\frac{1}{2}\right)\pi^{1/2}2^v v!}{\alpha\pi^{1/2}2^v v!} = \left(v+\tfrac{1}{2}\right)\alpha^2 \qquad \alpha^2 = (\hbar^2/mk_f)^{1/2}$$

$$\Rightarrow \quad \langle x^2\rangle_v = \left(v+\tfrac{1}{2}\right)\frac{\hbar}{(mk_f)^{1/2}}$$

位移随量子数增大而增大



平均势能正比于位移平方期望

$$\langle V\rangle_v = \tfrac{1}{2}\langle k_f x^2\rangle_v = \tfrac{1}{2}k_f\langle x^2\rangle_v = \tfrac{1}{2}\left(v+\tfrac{1}{2}\right)\hbar\left(\frac{k_f}{m}\right)^{1/2}$$

$$\langle V\rangle_v = \tfrac{1}{2}\left(v+\tfrac{1}{2}\right)\hbar\omega = \tfrac{1}{2}E_v$$

$$\langle E_k\rangle_v = E_v - \langle V\rangle_v = E_v - \tfrac{1}{2}E_v = \tfrac{1}{2}E_v$$

动能与势能各占一半

If the potential energy of a particle has the form $V = ax^b$, then its mean potential and kinetic energies are related by

$$2\langle E_k\rangle = b\langle V\rangle \qquad \text{Virial theorem} \qquad (7E.14)$$

位力定理

简谐波函数的隧穿　　经典禁阻区域与转折点　　$E_v = \tfrac{1}{2}k_f x_{tp}^2$

基态隧穿概率为16%　　$x_{tp} = \pm\left(\frac{2E_v}{k_f}\right)^{1/2}$

习题：计算各态隧穿概率

问题三　转动

复习1：经典力学中的转动　　每个转动模式对应一个转动轴

二维转动平面内：圆周运动

角速度ω (angular velocity)

角动量J (angular momentum)

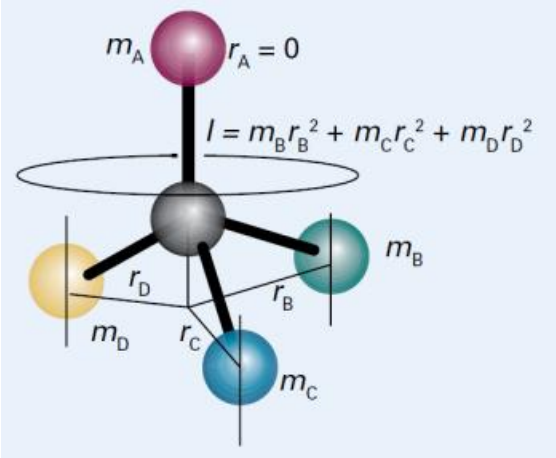
转动惯量I (moment of inertia)

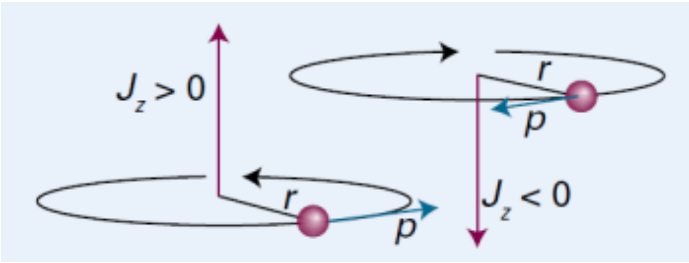
$$J = mvr = mr^2\omega = I\omega$$

对于多个质点

$$I = \sum_i m_i r_i^2$$

r_i是到转轴的距离





考虑转动的方向性，定义

$$\boldsymbol{J} = \boldsymbol{r} \times \boldsymbol{p}$$

$$J_x = yp_z - zp_y$$

$$J_y = zp_x - xp_z$$

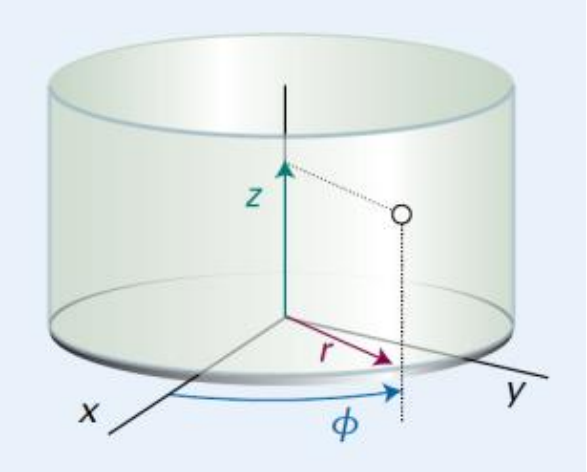
$$J_z = xp_y - yp_x$$

转动动能

$$E_k = \frac{1}{2} I \boldsymbol{\omega}^2 = \frac{\boldsymbol{J}^2}{2I}$$

复习2：柱坐标系与球坐标系

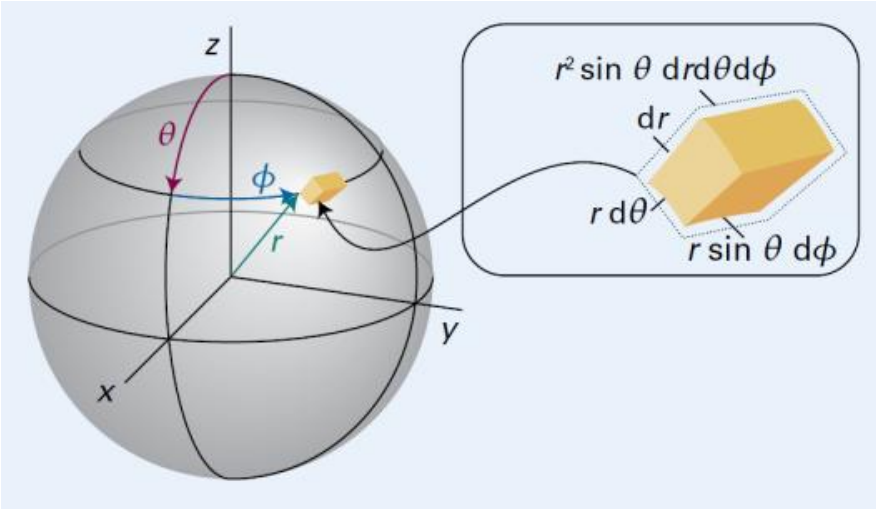
cylindrical coordinates

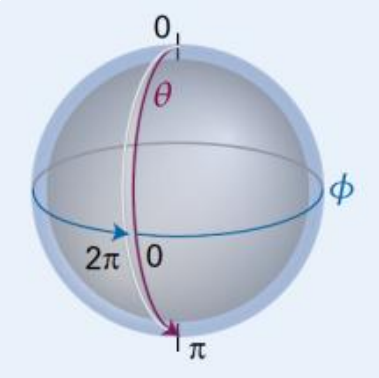


$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial}{\partial r}) + \frac{1}{r^2}\frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2}$$

$$dxdydz = rdrd\phi dz$$

spherical coordinates





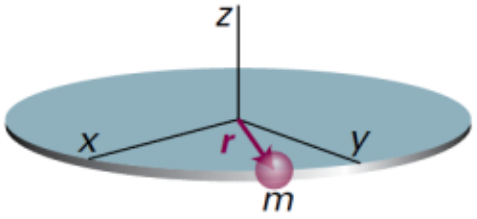
经度与纬度

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\frac{\partial}{\partial \theta}) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial \phi^2}$$

$$dxdydz = r^2\sin\theta drd\theta d\phi$$

θ：azimuth 地平经度
ϕ：colatitude 余纬度

二维转动



薛定谔方程限制在二维平面上半径为r的圆周

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\psi(x,y)=E\psi(x,y)$$

$$\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}=\frac{\partial^2}{\partial r^2}+\frac{1}{r}\frac{\partial}{\partial r}+\frac{1}{r^2}\frac{\partial^2}{\partial \phi^2}$$

ψ只是ϕ的函数，从而

$$-\frac{\hbar^2}{2mr^2}\frac{\mathrm{d}^2\psi(\phi)}{\mathrm{d}\phi^2}=E\psi(\phi)$$

$$-\frac{\hbar^2}{2I}\frac{\mathrm{d}^2\psi(\phi)}{\mathrm{d}\phi^2}=E\psi(\phi)$$

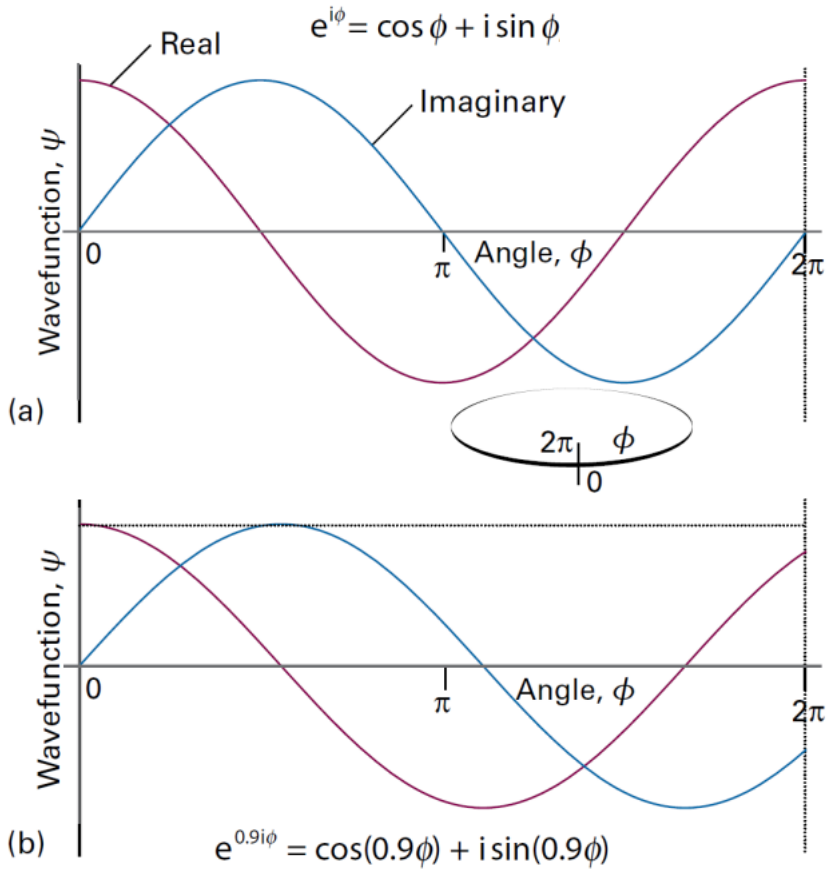
$$\psi(\phi+2\pi)=\psi(\phi)$$

注意：周期性边界条件导致量子化！

解得

$$\psi_{m_l}(\phi)=\frac{e^{im_l\phi}}{(2\pi)^{1/2}}$$

$$E_{m_l}=\frac{m_l^2\hbar^2}{2I}\quad m_l=0,\ \pm1,\ \pm2,\ldots$$



$$e^{im_l\phi},\ e^{-im_l\phi},\ \cos m_l\phi,\ \sin m_l\phi$$

能量简并，都是本征态

轨道角动量$\boldsymbol{l}$

经典力学 $$l_z = xp_y - yp_x$$

量子力学 $$\hat{l}_z = \frac{\hbar}{\mathrm{i}}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$

用柱坐标改写

$$\hat{l}_z = \frac{\hbar}{\mathrm{i}}\frac{\mathrm{d}}{\mathrm{d}\phi}$$

验证本征函数

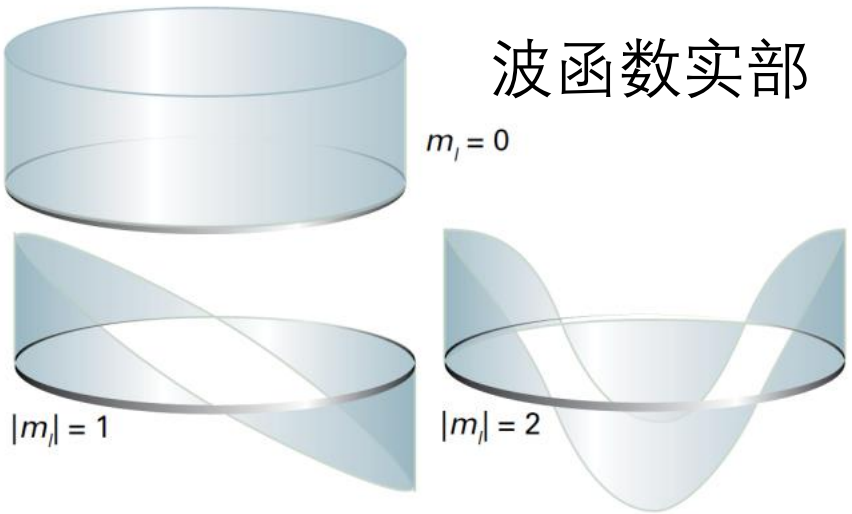
$$\hat{l}_z\psi_{m_l} = \frac{\hbar}{\mathrm{i}}\frac{\mathrm{d}}{\mathrm{d}\phi}\mathrm{e}^{\mathrm{i}m_l\phi} = \frac{\hbar}{\mathrm{i}}\mathrm{i}m_l\overbrace{\mathrm{e}^{\mathrm{i}m_l\phi}}^{\psi_{m_l}} = m_l\hbar\psi_{m_l}$$

动能与角动量共用一组本征波函数 $\psi_{m_l}(\phi) = \frac{\mathrm{e}^{\mathrm{i}m_l\phi}}{(2\pi)^{1/2}}$ 位置(ϕ)与角动量不能同时测准

$$\hat{l}_z\psi_{m_l}(\phi) = m_l\hbar\psi_{m_l}(\phi) \quad m_l = 0, \pm1, \pm2, \ldots$$

$\cos m_l\phi,\ \sin m_l\phi$ 是角动量本征态的叠加态

波函数实部



- 能量、角动量量子化
- 量子数越高，波函数振荡波长越短，能量、角动量越大
- 除基态外，各能态二重简并，对应**顺/逆时针旋转**
- 没有零点能：基态粒子不动

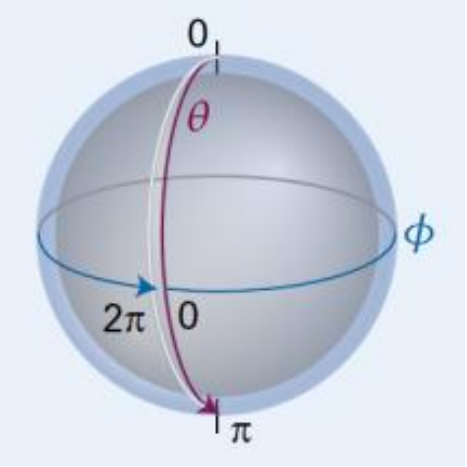
三维转动

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

改写Laplacian

Legendrian

$$\nabla^2 = \frac{1}{r}\frac{\partial^2}{\partial r^2}r + \frac{1}{r^2}\Lambda^2 \qquad \Lambda^2 = \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2} + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta}$$



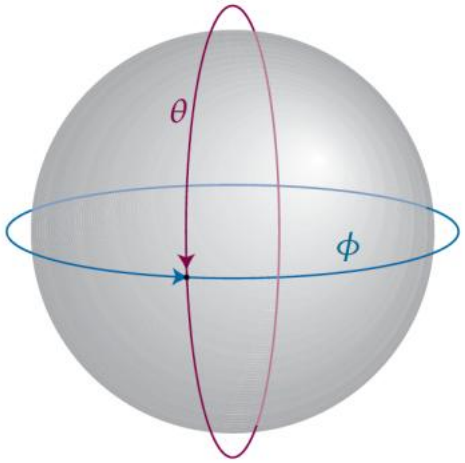
薛定谔方程退化为在二维球面上的运动方程

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\Lambda^2\psi(\theta,\phi) = E\psi(\theta,\phi)$$

$$-\frac{\hbar^2}{2I}\Lambda^2\psi(\theta,\phi) = E\psi(\theta,\phi)$$

球面对应两个边界条件

因此需引入两个量子数



解得 $\Lambda^2 Y_{l,m_l}(\theta,\phi) = -l(l+1)Y_{l,m_l}(\theta,\phi)$,

$l = 0, 1, 2, \ldots \quad m_l = 0, \pm 1, \ldots, \pm l$

本征态是球谐函数(spherical harmonics)

$$E_{l,m_l} = l(l+1)\frac{\hbar^2}{2I} \quad l=0, 1, 2\ldots \quad m_l = 0, \pm 1, \ldots \pm l$$

球谐函数同时是角动量z方向的本征态

$$\hat{l}_z Y_{l,m_l}(\theta,\phi) = \hat{l}_z \overbrace{e^{im_l\phi} f(\theta)}^{Y_{l,m_l}(\theta,\phi)} = \frac{\hbar}{i}\frac{\partial}{\partial\phi} e^{im_l\phi} f(\theta) = m_l\hbar \times e^{im_l\phi} f(\theta)$$
$$= m_l\hbar \times Y_{l,m_l}(\theta,\phi)$$

角动量z方向在球坐标与柱坐标中形式相同

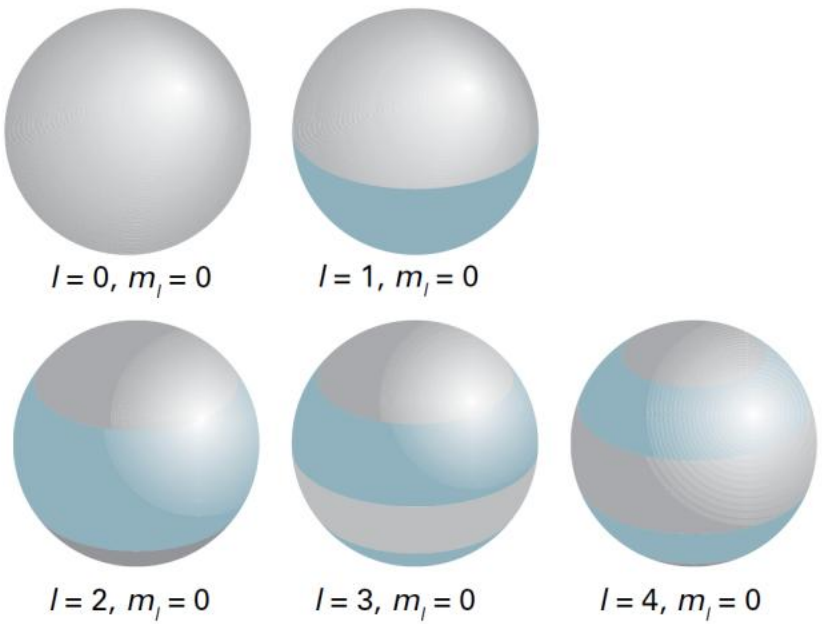
注意：z方向是人为指定的某个方向，可以是任意方向

l	m_l	$Y_{l,m_l}(\theta,\phi)$
0	0	$\left(\frac{1}{4\pi}\right)^{1/2}$
1	0	$\left(\frac{3}{4\pi}\right)^{1/2}\cos\theta$
	± 1	$\mp\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta e^{\pm i\phi}$
2	0	$\left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta - 1)$
	± 1	$\mp\left(\frac{15}{8\pi}\right)^{1/2}\cos\theta\sin\theta e^{\pm i\phi}$
	± 2	$\left(\frac{15}{32\pi}\right)^{1/2}\sin^2\theta e^{\pm 2i\phi}$
3	0	$\left(\frac{7}{16\pi}\right)^{1/2}(5\cos^3\theta - 3\cos\theta)$
	± 1	$\mp\left(\frac{21}{64\pi}\right)^{1/2}(5\cos^2\theta - 1)\sin\theta e^{\pm i\phi}$
	± 2	$\left(\frac{105}{32\pi}\right)^{1/2}\sin^2\theta\cos\theta e^{\pm 2i\phi}$
	± 3	$\mp\left(\frac{35}{64\pi}\right)^{1/2}\sin^3\theta e^{\pm 3i\phi}$

l：角量子数

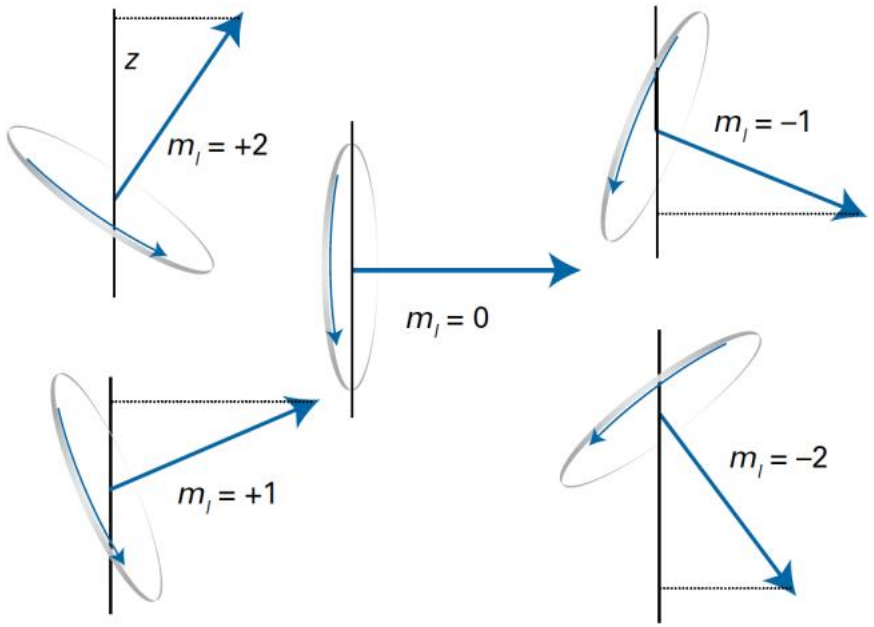
m_l：磁量子数

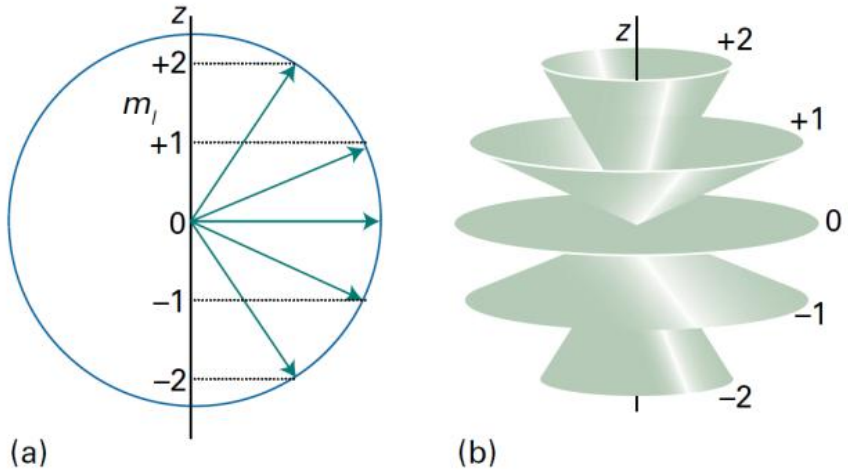
- 能量、角动量、角动量分量量子化
- 角量子数越高，角动量越大，能量越大
- 磁量子数绝对值越高，角动量z分量越大
- 能量与m_l无关，因为不依赖于旋转方向
- 各能态(2l+1)重简并，即有**(2l+1)个旋转方向**
- 没有零点能：基态粒子不动
- l越大，θ波函数振荡越强，节点越多



磁量子数为零时波函数的节线

轨道角动量与空间量子化





x/y与z分量不能同时测准，
亦即圆锥的方向不明

轨道角动量各分量

$$\hat{l}_x = \frac{\hbar}{\mathrm{i}}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)$$

$$\hat{l}_y = \frac{\hbar}{\mathrm{i}}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right)$$

$$\hat{l}_z = \frac{\hbar}{\mathrm{i}}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$

对易关系

$$[\hat{l}_x, \hat{l}_y] = i\hbar\hat{l}_z$$

$$[\hat{l}_y, \hat{l}_z] = i\hbar\hat{l}_x$$

$$[\hat{l}_z, \hat{l}_x] = i\hbar\hat{l}_y$$

$$[\hat{\boldsymbol{l}}^2, \hat{l}_q] = 0, \quad q = x, y, z$$

思考：如何简单证明？

角动量模方与任一分量
对易，有共同本征态

任意两分量均不对易，
没有共同本征态

利用转动能级差测键长

HI (I-127)的键长为160pm，问转动基态与第一激发态间发生跃迁吸收电磁波频率是多少？

答：由于I很重，可以假设I不动，H绕I转动（不考虑振动），从而转动惯量

$$I=\overbrace{(1.675\times10^{-27}\,\mathrm{kg})}^{m_{^1\mathrm{H}}}\times\overbrace{(1.60\times10^{-12}\,\mathrm{m})^2}^{R^2}=4.29\times10^{-47}\,\mathrm{kg\,m^2}$$

如已知吸收频率和转动惯量，可以反推键长，可以精确测量

$$\frac{\hbar^2}{2I}=\frac{(1.055\times10^{-34}\,\mathrm{Js})^2}{2\times(4.29\times10^{-47}\,\mathrm{kg\,m^2})}=1.30\times10^{-22}\,\mathrm{J}$$

$$\nu=\frac{\Delta E}{h}=\frac{2.60\times10^{-22}\,\mathrm{J}}{6.626\times10^{-34}\,\mathrm{Js}}=3.92\times10^{11}\,\overbrace{\mathrm{s^{-1}}}^{\mathrm{Hz}}=392\,\mathrm{GHz}$$

$$\lambda=\frac{c}{\nu}=\frac{2.998\times10^{8}\mathrm{m\ s^{-1}}}{3.92\times10^{11}\mathrm{s^{-1}}}=0.765\mathrm{mm}$$

微波波长